

OCR A-Level Computer Science (H446)
Computer Systems (Component 01)
Topic 1.1: The characteristics of contemporary processors,
input, output and storage devices

Mock Examination Paper

INSTRUCTIONS TO CANDIDATES

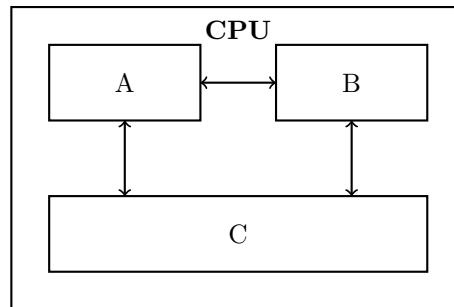
- Answer **ALL** questions.
- The total number of marks for this paper is 40.
- The recommended time for this paper is 45 minutes.
- Write your answers in the spaces provided.
- The quality of your written communication will be assessed in your answers.

Grade	Mark Range
A*	36-40
A	32-35
B	28-31
C	24-27
D	20-23
E	16-19

Question 1

1. Define the term 'Von Neumann Architecture' and explain two key features that distinguish it from Harvard Architecture. [4 marks]

2. The diagram below shows a simplified structure of a CPU:



- (a) Identify components A, B, and C. [3 marks]
 - (b) Explain the role of component A in the Fetch-Decode-Execute cycle. [3 marks]
3. Explain what is meant by the term 'clock speed' and how it affects the performance of a processor. [3 marks]

Question 2

4. Describe the purpose of the following CPU registers: [4 marks]
 - (a) Program Counter (PC)

 - (b) Current Instruction Register (CIR)

5. A processor has a clock speed of 2.4 GHz.
 - (a) Calculate how many clock cycles this processor can execute in one second. Show your working. [2 marks]

 - (b) If an instruction takes 3 clock cycles to complete, calculate how many instructions this processor can execute per second. Show your working. [2 marks]

6. Explain how each of the following affects processor performance: [4 marks]
 - (a) Cache size

 - (b) Number of cores

Question 3

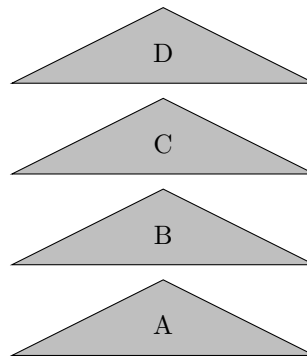
7. Compare CISC and RISC processor architectures by explaining: [4 marks]

(a) One key difference in their instruction sets

(b) One advantage and one disadvantage of RISC compared to CISC

8. Explain what is meant by 'pipelining' in CPU design and how it improves processor performance. [3 marks]

9. The following diagram represents a typical memory hierarchy in a computer system:



Identify what each level (A, B, C, and D) represents in a typical memory hierarchy, and explain why this hierarchical structure is used. [4 marks]

Question 4

10. For each of the following, state whether it is primarily an input device, output device, or both, and explain your answer: [4 marks]

(a) Touchscreen

(b) Network Interface Card (NIC)